

Anemia - Factoid Sheet

Week 7 Hematology • Lecture 4 • Dalgetty • memorize-first companion

ScholarRx: [Anemias: Foundations & Frameworks](#) • [Anemias: Putting It All Together](#)

Algorithm cold: Low Hb -> CBC/diff -> **retic** (marrow response?) -> **MCV** -> smear + targeted labs -> identify the cause.

RBC indices

- **MCV:** <80 micro, 80-100 normo, >100 macro.
- **RDW:** size variation; often high in IDA.
- **Retic:** high = loss/destruction if marrow intact; low/inadequate = production problem.
- A "normal" retic % during anemia can be inappropriately low.

Hepcidin

Hepcidin binds/degrades ferroportin.

- High hepcidin -> less gut iron absorption.
- High hepcidin -> macrophages retain iron.
- Inflammation -> high hepcidin -> functional iron restriction.

Microcytic anemias - iron patterns

	Fe	Ferritin	TIBC	Sat	Clue
IDA	down	down	up	down	Pica, ice, koilonychia, bleeding
Inflammation	down	normal/up	down	down	High hepcidin, chronic disease
Thal trait	normal	normal/up	normal	normal	Very low MCV, target cells, high A2 in beta trait
Sideroblastic	up	up	variable	up	Ring sideroblasts, iron trapped in mitochondria

IDA associations

- Ice craving/pica
- Restless legs
- Koilonychia
- Plummer-Vinson = IDA + esophageal web/dysphagia
- Always find source: menses, GI, pregnancy, GU, malabsorption, donation, parasite

Sideroblastic / lead

- Prussian blue **ring sideroblasts**.
- Acquired: alcohol, MDS, copper deficiency; B6-responsive subset.
- Lead: abdominal colic + neuropathy + **basophilic stippling** + Burton line.
- Lead inhibits ALA dehydratase + ferrochelatase.

Normocytic shortcuts

Retic	Think	Examples
High	Loss/destruction	Bleeding; hemolysis (LDH up, indirect bilirubin up, haptoglobin down)
Low	Underproduction	CKD/EPO deficiency, inflammation, marrow suppression, mixed disease

Transfusion: modern restrictive approach in many *stable* adults = consider around Hb <7 g/dL; ~8 g/dL may be used in orthopedic surgery/preexisting CVD. Active bleeding/shock/symptoms override a simplistic number.

Macrocytic + Diagnostic Factoids

Megaloblastic vs non-megaloblastic

Megaloblastic	Non-megaloblastic
Impaired DNA synthesis; macro-ovalocytes + hypersegmented neutrophils ; B12, folate, DNA-synthesis drugs.	Alcohol, liver disease, hypothyroidism, reticulocytosis, some marrow disorders; no classic hypersegmented-neutrophil pattern.

B12 vs folate

	B12	Folate
Homocysteine	up	up
MMA	up	normal
Neurologic	yes - vibration/position loss, paresthesia, ataxia, etc.	No classic B12-type neuro syndrome
Stores	Years	Months
Smear	Macrocytes/macro-ovalocytes + hypersegmented neutrophils	

Pernicious anemia

Autoimmune intrinsic-factor/parietal-cell problem -> B12 malabsorption.

- Chronic atrophic gastritis
- Other autoimmune disease association
- Increased gastric carcinoma risk
- Parenteral or sufficiently high-dose oral B12 can treat

Do not mask B12 deficiency: folate alone can improve the anemia while neurologic injury continues.

If you see X -> think Y

• Ice craving + spoon nails -> IDA • Low Fe + high ferritin + low TIBC -> inflammation • CKD + low retic -> inadequate EPO • Jaundice + acute Hb fall -> order LDH/haptoglobin/bilirubin + smear	• Macrocytes + hypersegmented neutrophils -> megaloblastic • High MMA + neuro signs -> B12 deficiency • Basophilic stippling + abdominal/neuro symptoms -> lead • Ring sideroblasts -> sideroblastic anemia
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Cost-effective treatment logic

Problem	First logic
IDA	Oral ferrous salt when tolerated/absorbed + find/treat source; IV iron selectively.
Inflammation	Treat underlying disorder; iron only when true coexisting deficiency exists.
CKD	Correct iron deficiency; ESA decisions individualized; current KDIGO avoids high Hb targets.
B12	Replace B12 (IM or high-dose oral depending setting/cause).
Folate	Folic acid replacement - after excluding/covering B12 when clinically relevant.

Last 5 minutes: Retic tells production vs loss/destruction. MCV narrows morphology. IDA ferritin down/TIBC up; inflammation ferritin up/TIBC down. B12 = MMA up + neuro. Folate = MMA normal. Lead = stippling. Sideroblastic = ringed marrow iron.

Primary source: Dalgetty 62-slide Approach to Anemia deck. Official Week 7 LOs set the emphasis. Current AABB/KDIGO items are external guideline enrichment.